

IoT-based Autonomous Disaster Rescue Robot with Human Detection and Obstacle Avoidance

Project Report

Submitted towards the partial fulfillment of the requirements for
the degree of
Bachelor of Engineering

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SUPERVISOR'S CERTIFICATE

This is to certify that the work reported in the project entitled “IoT-based Autonomous Disaster Rescue Robot with Human Detection and Obstacle Avoidance”, submitted by **Aditya Kumar (23010104005), Ansh Lohia (23010104011), Anshdeep Singh (23010104012), and Lakshay Sharma (23010104039)** to **H.P. Technical University, Hamirpur**, is a bonafide record of original work carried out by the under our supervision. This work has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma.

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DECLARATION

We hereby declare that the work presented in this project entitled “*IoT-based Autonomous Disaster Rescue Robot with Human Detection and Obstacle Avoidance*” is an original and authentic record of our work carried out under the supervision of **Er. Pooja Sharma** at **J.N. Government Engineering College, Sundernagar**. We further declare that this work has not been submitted ,either in part or in full,to any other university or institution for the award of any degree ,diploma , or certificate. We take full responsibility for the contents of this work.

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Abstract

This project presents an IoT-based autonomous disaster rescue robot designed to detect human presence and safely navigate hazardous environments. Using an ESP32 microcontroller integrated with CO₂, ultrasonic, and PIR sensors, the robot identifies signs of human life and avoids obstacles in debris-filled disaster zones. Sensor data is transmitted wirelessly via Bluetooth for real-time monitoring, enabling rescue teams to locate trapped survivors efficiently while significantly reducing risk to human responders.

The proposed system is implemented using an ESP32 38-pin WROOM module interfaced with dual HC-SR04 ultrasonic sensors for distance measurement and obstacle avoidance, an SCD40 CO₂ sensor for human presence detection based on exhaled carbon dioxide concentration, an L298 motor driver for four-wheel drive control, and a buzzer with LED for alert signaling. The robot is controlled remotely over Bluetooth and continuously acquires sensor data, processing it in real time to enable autonomous decision-making.

The system detects human presence by monitoring CO₂ concentration levels above 1700 ppm, a threshold indicative of human exhalation in enclosed or debris-filled spaces. Obstacle detection is achieved when the front ultrasonic sensor measures a distance of less than 15cm. Upon detection of either condition, the system triggers audible and visual alerts and transmits data to the operator's device via Bluetooth.

The developed system is expected to demonstrate effective obstacle avoidance in cluttered and low-visibility disaster environments. Real-time data transmission via the ESP32 Bluetooth module is expected to provide continuous situational awareness to rescue teams, thereby improving response time, operational safety, and the overall effectiveness of disaster rescue-missions.

Chapter 1

Introduction

Disaster rescue operations are among the most challenging and time-critical activities faced by emergency response teams. Hazardous conditions including structural collapse, toxic gas accumulation, poor visibility, and the threat of secondary disasters create an environment that is extremely dangerous for human rescuers. Rapid identification and location of trapped survivors is critical to saving lives, yet the constraints of the environment often delay or prevent effective search operations.

Robotic systems offer a compelling solution to these challenges. An autonomous rescue robot capable of navigating debris-filled environments, detecting signs of human presence, and transmitting real-time information to rescue teams can dramatically improve search efficiency while reducing risk to responders. The integration of IoT technologies enables remote monitoring and control, making such systems both practical and scalable.

Figure 1.1 illustrates the generalized system overview of the proposed project, showing the major components and their interaction in scenario.

Project Overview

This project focuses on designing and implementing an IoT -based autonomous disaster rescue robot that can locate survivors in hazardous disaster zones. The system integrates a CO₂ sensor and dual ultrasonic sensors with an ESP32 microcontroller for real-time human detection ,obstacle avoidance, and wireless monitoring over Bluetooth. The robot is mounted on a four-wheel smart chassis driven by DC motors via an L298 motor driver, and is capable of being controlled remotely while simultaneously broadcasting environ-mental sensor data. The system improves search efficiency, reduces response time, and enhances the safety of rescue teams operating in post-disaster environments.

The robot is compact, cost-effective, and energy-efficient, making it suitable for deployment in narrow and unstable environments. Its autonomous navigation capability reduces human effort and minimizes the risk faced by rescue personnel.

The system improves search efficiency, reduces response time, and enhances the safety of rescue teams operating in post-disaster environments. The project demonstrates the practical application of embedded systems, IoT, automation, and sensor integration in real-world disaster management and humanitarian rescue operations.

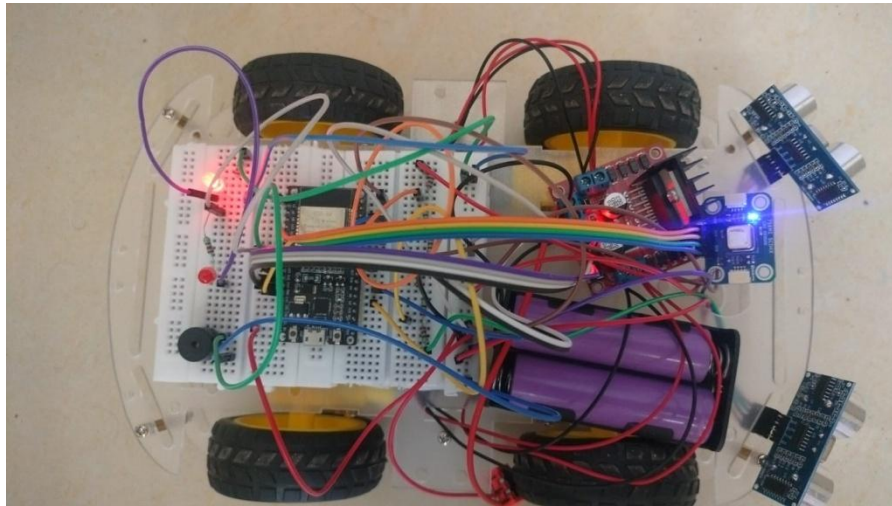


Figure1.1:Prototype of the IoT-based Autonomous Disaster Rescue Robot

Motivation

Disaster zones are highly dangerous and time-critical environments. Manual rescue operations put responders at significant risk and often delay victim identification. Existing robotic solutions either rely on single-sensor approaches that lack robustness, or are prohibitively expensive for widespread deployment.

This project is motivated by the need to develop an affordable, multi-sensor autonomous rescue robot that can safely navigate debris, accurately detect human presence through CO₂ concentration monitoring and proximity sensing, and provide real-time information to rescue teams for quicker and safer response.

Objectives

The key objectives of this project are:

- To design and fabricate a four-wheel autonomous rescue robot using the ESP32 microcontroller.
- To integrate an SCD40 CO₂ sensor for detecting human presence based on exhaled carbon dioxide levels.
- To implement obstacle detection and avoidance using dual HC-SR04 ultrasonic sensors.
- To enable real-time wireless communication and remote control via Bluetooth.
- To provide alert signaling through a buzzer and LED upon detection of humans or obstacles.

- To transmit continuous sensor data (CO₂,temperature,humidity,distance) to a remote monitoring device.

Scope of the Project

The project covers the hardware design, firmware development, and functional testing of an IoT-enabled rescue robot prototype. The scope includes sensor integration, motor control, Bluetooth-based remote operation, and real-time alert generation. The system is designed as a proof-of-concept for deployment in post-disaster search and rescue scenarios.

This project can be used in disaster situations such as earthquakes, landslides and collapsed buildings where human rescue operations are difficult and risky. The robot helps reduce rescue time, improves victim detection accuracy, and enhances the safety of rescue personnel.

Organization of the Report

This report is organized as follows. Chapter 2 presents a view of related literature on disaster rescue robotics. Chapter 3 describes the proposed methodology, system architecture, and algorithm. Chapter 4 presents the results and discussion. Chapter 5 concludes the report and outlines future scope.

Chapter 2

Literature Review

Recent studies on disaster rescue robotics highlight the effectiveness of autonomous mobile robots in hazardous environments. Researchers have demonstrated that integrating ultrasonic sensors for obstacle avoidance with gas and environmental sensors for human detection improves survivor identification in low-visibility conditions. The use of IoT-enabled microcontrollers such as the ESP32 enables real-time data transmission and remote monitoring. Multi-sensor fusion combined with wireless connectivity enhances the reliability, safety, and efficiency of robotic search and rescue operations.

Overview of Existing Work

Existing disaster rescue systems primarily rely on manual search operations, trained rescue teams, and basic robotic platforms with limited sensing capabilities. Some earlier robots use single sensors such as cameras, ultrasonic sensors, or gas detectors, which restrict accurate human detection in complex environments. While a few IoT-based solutions provide remote monitoring, they often lack autonomous navigation, multi-sensor integration, and real-time decision-making.

A major challenge in the field is the cost and complexity of systems. High-performance rescue robots such as those developed for DARPA Robotics Challenge competitions involve sophisticated hardware and software stacks that are not feasible for rapid low-cost deployment. There is therefore a well-recognized need for affordable, practical multi-sensor autonomous rescue robots for disaster response.

The following subsections summarize selected studies from journals, conference papers, and technical reports relevant to this project.

Review of Selected Literature

Study1:Multi-Sensor Rescue Robot

Pushpa Latha et al. [1] proposed a robot designed for rescue operations in disaster environments, focusing on multi-sensor integration and experimental validation. The system employed ultrasonic and infrared sensors for obstacle avoidance and environmental monitoring. The approach improves safety and navigation performance, but the robot has limited autonomy in highly cluttered areas due to the absence of gas-based human detection.

Study2:PIR-Based Human Detection Robot

Narasimha Rao[2] developed a low-cost human detection robot using a PIR (Passive Infrared) sensor. The system detects human presence based on body heat and motion, making it simple and cost-effective. However, the system shows limited accuracy in complex, real-world conditions where ambient heat sources or obstructions may trigger false positives or prevent detection of stationary survivors.

Study3:DARPA/NSF Study on Human-Robot Interaction in SAR

Carlson et al. [3] presented a comprehensive technical report on human-robot interaction in search and rescue operations, emphasizing the requirements, design considerations, and performance evaluation of robotic systems in disaster response. The report outlines key challenges in autonomous navigation, sensor performance, and human-robot coordination. While it provides foundational guidelines for robot design and deployment, the report does not focus on specific low-cost sensor implementations or IoT integration for real-time remote monitoring.

Literature Comparison

To better understand the differences between existing approaches, a comparative analysis is presented. Table 2.1 compares selected literature based on methodology, key features, and limitations.

Technique-Based Comparison

A technique-oriented comparison is shown in Table 2.2, highlighting tools, evaluation parameters, and outcomes used in previous studies.

Table2.1:Comparison of Existing Literature

Ref.	Study	Key Features	Limitations
[1]	Multi-sensor rescue robot	Obstacle avoidance, Experimental validation	Low autonomy in cluttered areas
[2]	Low-cost PIR Human detection	Simple, cost-effective, motion-based detection	Limited accuracy in real-world conditions
[3]	SAR robot design guidelines	Comprehensive design And evaluation framework	No IoT/low-cost sensor focus

Table2.2:Technique-BasedLiteratureComparison

Author	Tools/Techniques	Criteria	Outcome
Pushpa Latha Et al.	Multi-sensor rescue robot(ultrasonic,IR)	Accuracy, Safety	Improved navigation And safety
Narasimha Rao	PIR-based humande-Tection robot	Cost, Simplicity	Effective but limited accuracy
Carlson et al.	SAR robot design guidelines and HRI study	Reliability	Stable design framework

Research Gap and Motivation

Most existing disaster rescue robots either focus on basic human detection or simple navigation, and lack an integrated approach combining multi-sensor human detection, reliable obstacle avoidance, and real-time IoT-based wireless monitoring .Low-cost systems often perform poorly in complex, cluttered environments, creating a gap for a practical and scalable solution.

In particular, CO₂-based human detection – which exploits the elevated CO₂ concentration in exhaled breath–has not been widely explored in low-cost rescue robots. Unlike PIR sensors, CO₂ sensors such as the SCD40 can detect stationary survivors who are not generating detectable heat or motion, offering a complementary and robust detection mechanism. This project addresses this gap by integrating CO₂ sensing with ultrasonic obstacle avoidance and real-time Bluetooth communication in an affordable ESP32-based platform.

Chapter 3

Methodology

The robot continuously senses the environment using multiple sensors, processes fused data for human identification and safe path planning, and is guided by Bluetooth commands from an operator. Key events and sensor readings are transmitted wirelessly to a remote monitoring device via the ESP32's built in Bluetooth library.

Overall Methodology

The robot integrates an SCD40 CO₂ sensor and dual HC-SR04 ultrasonic sensors with an ESP32 microcontroller to detect human presence and obstacles. Sensor data is processed in real time for decision-making, enabling autonomous alert generation and operator-guided navigation with obstacle detection. Real-time Bluetooth communication transmits sensor readings and alerts to the rescue team for timely response.

Figure3.1 presents the flow chart of the proposed methodology.

System Architecture

The system consists of an ESP32 WROOM38- pin microcontroller interfaced with environmental and proximity sensors, motor drivers, and communication modules. The front HC-SR04 ultrasonic sensor measures the distance to obstacles ahead of the robot, while the right-side HC-SR04 monitors lateral clearance. The SCD40 sensor is connected via I²C (SDA: GPIO 21, SCL: GPIO 22) and provides CO₂ concentration, temperature, and relative humidity readings every four seconds. The L298 motor driver controls four DC motors through PWM signals from the ESP32. An LED and buzzer provide local alerts.

Table3.1 lists the primary hardware components and their respective functions.

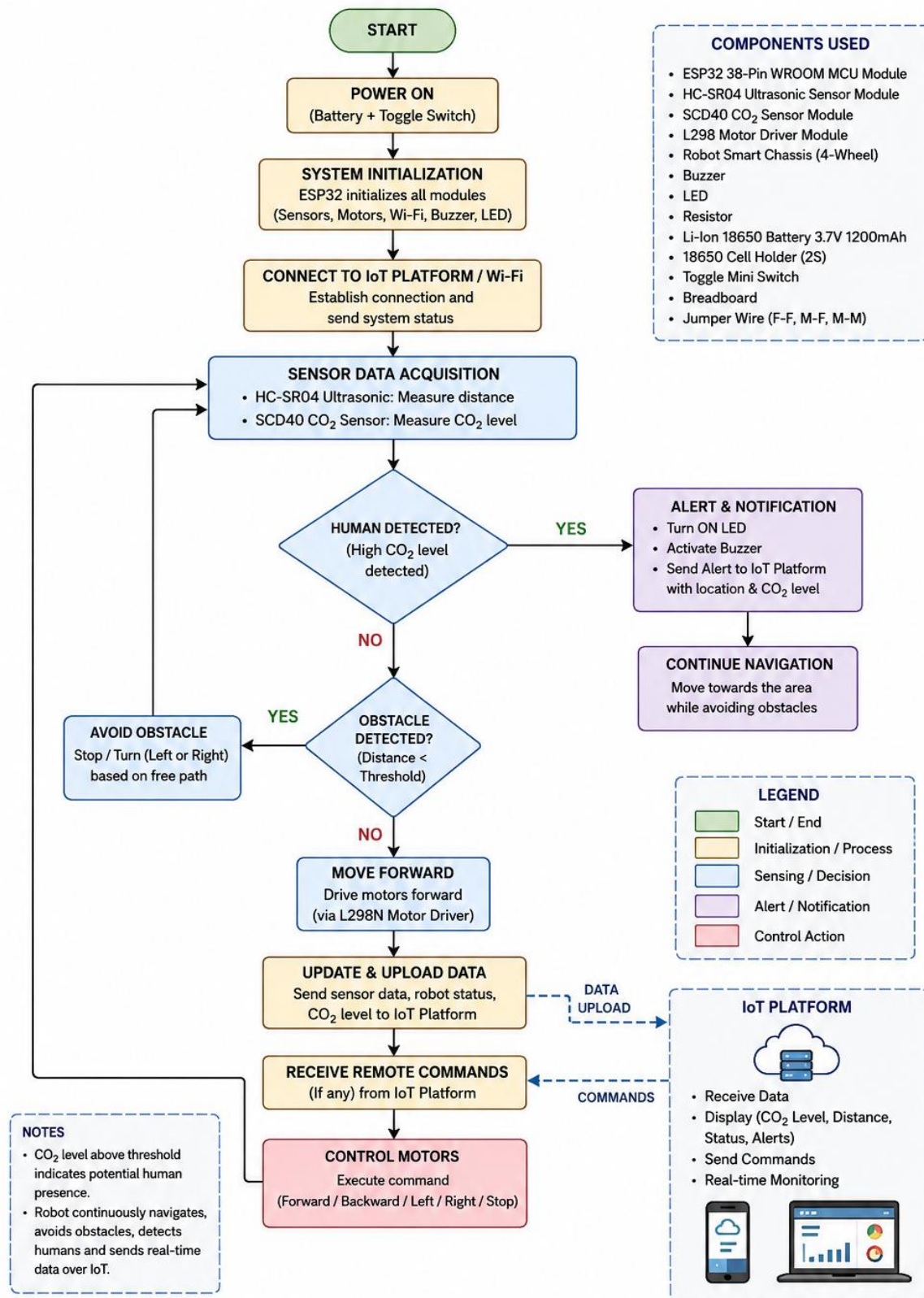


Figure3.1:Flow chart of the proposed methodology

Table3.1: System Architecture Components

Component	Function
ESP32 WROOM 38-Pin Module	Main microcontroller ;processes sensor data, controls motors, manages Bluetooth communication
HC-SR04 Ultrasonic Sensor (Front)	Measures distance to obstacles in front of the robot
HC-SR04 Ultrasonic Sensor (Right)	Monitors lateral clearance on the right side
SCD40CO ₂ Sensor	Detects human presence via CO ₂ concentration; also measures temperature and humidity
L298 Motor Driver Module	Drives DC motors for forward ,backward, left, and right movement
DC Motors(4×)	Provide four-wheel drive for robot locomotion
Buzzer	Audible alert on human or obstacle detection
LED	Visual alert indicate or
Li-Ion 18650 Battery(2S)	Power supply for the entire system
Bluetooth (ESP32built-in)	Wireless communication with operator's device

The system architecture of the proposed IoT-based disaster rescue robot consists of an ESP32 microcontroller that acts as the central processing unit for controlling sensors, motors, and communication modules. The HC-SR04 ultrasonic sensor is used for obstacle detection and avoidance, while the SCD40 CO₂ sensor helps in detecting possible human presence in disaster areas.

The L298 motor driver controls the movement of the 4-wheel robotic chassis, enabling autonomous navigation. IoT connectivity through the ESP32 allows real-time monitoring, data transmission, and remote control using a wireless network. The buzzer and LED provide alert indications whenever a human is detected or an emergency situation occurs.

Hardware Components List

The complete bill of materials for the prototype is listed in Table 3.2.

Table3.2:Bill of Materials

S. No.	Component	Quantity
1	ESP323 8-Pin WROOM MCU Module	1
2	HC-SR 04 Ultrasonic Sensor Module	2
3	L298 Motor Driver Module	1
4	SCD40 CO ₂ Sensor Module	1
5	Robot Smart Chassis (4-Wheel)	1
6	Buzzer	1
7	LED	1
8	Resistor	As required
9	Li-Ion 18650 Battery 3.7 V 1200 mAh	2
10	18650 Cell Holder (2S)	1
11	Toggle Mini Switch	1
12	Bread board	1
13	Jumper Wire Female-to-Female	As required
14	Jumper Wire Male-to-Female	As required
15	Jumper Wire Male-to-Male	As required

Pin Configuration

The ESP 32 GPIO pin assignments used in the firmware are listed in Table 3.3.

Table3.3:ESP 32 GPIO Pin Configuration

GPIO Pin	Connected To	Function
GPIO26	L298IN1	Motor A–Forward
GPIO27	L298IN2	Motor A–Backward
GPIO14	L298IN3	Motor B–Forward
GPIO12	L298IN4	Motor B–Backward
GPIO5	HC-SR04(Front) TRIG	Front Ultrasonic Trigger
GPIO18	HC-SR04(Front) ECHO	Front Ultrasonic Echo
GPIO4	HC-SR04(Right) TRIG	Right Ultrasonic Trigger
GPIO2	HC-SR04(Right) ECHO	Right Ultrasonic Echo
GPIO21	SCD40SDA	I ² CData
GPIO22	SCD40SCL	I ² Cclock
GPIO33	LED	Visual Alert Output
GPIO25	Buzzer	Audible Alert Output

Proposed Algorithm

The robot initializes all sensors and communication modules on startup, then enters a continuous loop. In each loop iteration, Bluetooth commands are checked and motor directions updated immediately for responsive control. Every four seconds, sensor data is acquired from both ultra sonic sensors and the SCD40, and alert conditions are evaluated.

Algorithm: IoT Rescue Robot Operation

Step1: Initialization

Initialize ESP32, Serial, Bluetooth("ESP32ROBOT")
Configure GPIO pins for motors, ultrasonic sensors, LED, buzzer
Initialize I²C bus;
start SCD40 periodic measurement
Wait 5 seconds for SCD40 warm up ; stop motors

Step2: Bluetooth Command Processing (every loop iteration)

If Bluetooth data available:
Read command character
If 'F': run motors forward
If 'B': run motors backward
If 'L': turn left
If 'R': turn right
If 'S': stop motors

Step3: Sensor Reading (every 4 seconds)

Trigger front ultrasonic; compute front distance d_f
Trigger right ultrasonic; compute right distance d_r
Request SCD40 measurement; read CO₂(c), temperature(T), humidity(H)

Step4: Alert Decision

If $c > 1700$ ppm: activate LED and buzzer ; transmit "HUMAN DETECTED"
Else if $d_f < 15$ cm: deactivate LED and buzzer ; transmit "OBSTACLE DETECTED"
Else: deactivate LED and buzzer

Step5: Data Transmission

Transmit d_f, d_r, c, T , Hover Bluetooth and Serial

Step6: Repeat from Step2

END

Human Detection Principle

Human detection in this system is based on the concentration of CO₂ in the environment. Normal atmospheric CO₂ is approximately 400 ppm. Human exhalation raises local CO₂

concentration significantly, especially in enclosed or confined spaces. The SCD40 sensor measures CO₂ using photo acoustic spectroscopy. When the reading exceeds the detection threshold of 1800 ppm, the system flags human presence and triggers an alert.

The distance measured by the front HC-SR04 is computed as:

$$d = \frac{t_{echo} \times v_{sound}}{2} \quad (3.1)$$

Where t_{echo} is the echo pulse duration in micro seconds and $v_{sound} \approx 0.034 \text{ cm}/\mu\text{s}$.

CONCLUSION

This chapter presented the complete methodology for the IoT-based rescue robot, including the system architecture, hardware bill of materials, GPIO pin configuration, operational algorithm, and the principles of CO₂-based human detection and ultrasonic obstacle avoidance. The described methodology provides a clear and structured plan that was implemented in the prototype described in subsequent chapters.

Chapter4

Results & Discussion

Overview

This chapter presents the implementation results and observations from the prototype of the IoT -based autonomous disaster rescue robot. The system was assembled, programmed , and tested to evaluate its capability for human detection via CO₂ sensing, obstacle detection via ultrasonic measurement, and wireless communication via Blue-tooth.

Prototype Implementation

The robot prototype was assembled on a four-wheel smart chassis. The ESP32 WROOM module and all peripheral components were mounted on a breadboard secured to the chassis. The SCD40 CO₂ sensor was connected via I²C and the two HC-SR04 ultrasonic sensors were positioned at the front and right side of the chassis .The L298 motor driver was connected to four DC motors for loco motion .The system was powered by two Li-Ion 18650 cells in series configuration ,providing approximately 7.4 V through a toggle switch.

Figure 4.1 shows the assembled prototype of the rescue robot.

Firm ware and Communication

The firmware was developed in the Arduino framework for ESP32.The robot connects over Bluetooth Classic as “ESP32 ROBOT” and accepts single-character movement com-mands (F, B, L, R, S). Sensor data and alert messages are streamed to the connected device every four seconds. Table 4.1 lists the supported Bluetooth commands.

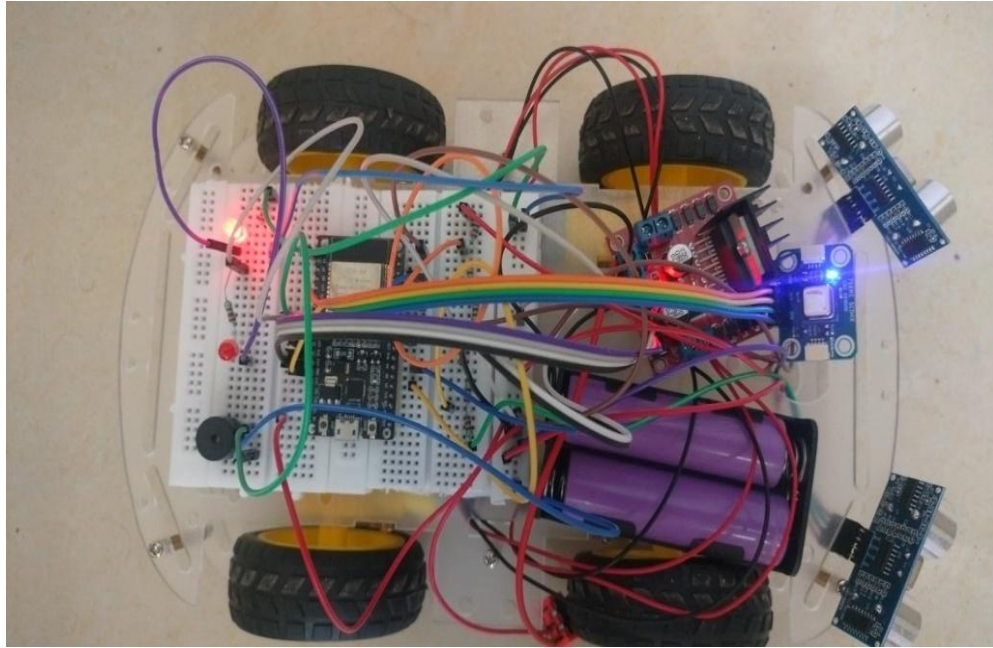


Figure4.1:Assembled prototype of the IoT-based Rescue Robot

Table4.1:Bluetooth Command Set

Command Character	Robot Action
F	Move Forward
B	Move Backward
L	Turn Left
R	Turn Right
S	Stop

ESP32 Firmware Listing

The complete firmware for the rescue robot is provided below.The code handles Bluetooth communication, motor control via PWM ,ultrasonic distance measurement ,SCD40 sensor reading over I²C, and alert generation.

Listing4.1:ESP32 Rescue Robot Firmware (ArduinoC++)

```
#include <Wire.h>
#include <BluetoothSerial.h>

BluetoothSerial SerialBT;

// ===== PIR SENSOR =====
#define PIR 16

// ===== MOTOR PINS =====
#define IN1 26
#define IN2 27
```

```

#define IN3 14
#define IN4 12

// ===== MOTOR SPEED =====
#define SPEED 200

// ===== ULTRASONIC SENSORS =====

// LEFT SENSOR
#define TRIG_L 5
#define ECHO_L 18

// RIGHT SENSOR
#define TRIG_R 4
#define ECHO_R 2

// CENTER SENSOR
#define TRIG_C 19
#define ECHO_C 23

// ===== LED & BUZZER =====
#define LED 33
#define BUZZER 25

// ===== SCD40 =====
#define SCD40_ADDR 0x62

uint16_t co2;
float temperature;
float humidity;

// ===== VARIABLES =====
char cmd;

bool autoMode = false;

unsigned long lastSensorTime = 0;

// =====
// MARIO STYLE TUNE
// =====

void playTune() {

    tone(BUZZER, 659);
    delay(150);

    tone(BUZZER, 659);
    delay(150);

    tone(BUZZER, 659);
    delay(150);

    tone(BUZZER, 523);
    delay(150);

    tone(BUZZER, 659);
    delay(150);

    tone(BUZZER, 784);
    delay(300);

```

```

tone(BUZZER, 392);
delay(300);

noTone(BUZZER);
}

// =====
// SCD40 FUNCTIONS
// =====

void sendCommand(uint16_t cmdData) {

    Wire.beginTransmission(SCD40_ADDR);

    Wire.write(cmdData >> 8);
    Wire.write(cmdData & 0xFF);

    Wire.endTransmission();
}

bool readData(uint16_t &co2, float &temp, float &hum) {

    Wire.requestFrom(SCD40_ADDR, 9);

    if (Wire.available() != 9)
        return false;

    uint8_t data[9];

    for (int i = 0; i < 9; i++) {
        data[i] = Wire.read();
    }

    co2 = (data[0] << 8) | data[1];

    uint16_t rawTemp = (data[3] << 8) | data[4];
    uint16_t rawHum = (data[6] << 8) | data[7];

    temp = -45 + 175 * ((float)rawTemp / 65535.0);
    hum = 100 * ((float)rawHum / 65535.0);

    return true;
}

// =====
// ULTRASONIC FUNCTION
// =====

int getDistance(int trigPin, int echoPin) {

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);

    digitalWrite(trigPin, LOW);

    long duration = pulseIn(echoPin, HIGH, 30000);

```

```

    int distance = duration * 0.034 / 2;

    if (distance <= 0 || distance > 400)
        distance = 400;

    return distance;
}

// =====
// MOTOR FUNCTIONS
// =====

void forward() {

    analogWrite(IN1, 0);
    analogWrite(IN2, SPEED);

    analogWrite(IN3, 0);
    analogWrite(IN4, SPEED);
}

void backward() {

    analogWrite(IN1, SPEED);
    analogWrite(IN2, 0);

    analogWrite(IN3, SPEED);
    analogWrite(IN4, 0);
}

void left() {

    analogWrite(IN1, SPEED);
    analogWrite(IN2, 0);

    analogWrite(IN3, 0);
    analogWrite(IN4, SPEED);
}

void right() {

    analogWrite(IN1, 0);
    analogWrite(IN2, SPEED);

    analogWrite(IN3, SPEED);
    analogWrite(IN4, 0);
}

void stopMotor() {

    analogWrite(IN1, 0);
    analogWrite(IN2, 0);

    analogWrite(IN3, 0);
    analogWrite(IN4, 0);
}

// =====
// SETUP
// =====

```

```

void setup() {

  Serial.begin(115200);

  SerialBT.begin("ESP32_ROBOT");

  // PIR
  pinMode(PIR, INPUT);

  // MOTOR PINS
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  // LEFT ULTRASONIC
  pinMode(TRIG_L, OUTPUT);
  pinMode(ECHO_L, INPUT);

  // RIGHT ULTRASONIC
  pinMode(TRIG_R, OUTPUT);
  pinMode(ECHO_R, INPUT);

  // CENTER ULTRASONIC
  pinMode(TRIG_C, OUTPUT);
  pinMode(ECHO_C, INPUT);

  // LED & BUZZER
  pinMode(LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);

  // I2C
  Wire.begin(21, 22);

  delay(1000);

  // START SCD40
  sendCommand(0x21B1);

  delay(5000);

  stopMotor();

  Serial.println("ESP32 ROBOT READY");

  SerialBT.println("ESP32 ROBOT READY");

  SerialBT.println("A = AUTO MODE");
  SerialBT.println("M = MANUAL MODE");

  SerialBT.println("F = FORWARD");
  SerialBT.println("B = BACKWARD");
  SerialBT.println("L = LEFT");
  SerialBT.println("R = RIGHT");
  SerialBT.println("S = STOP");
}

// =====
// LOOP
// =====

```

```

void loop() {

    // =====
    // BLUETOOTH CONTROL
    // =====

    while (SerialBT.available()) {

        cmd = SerialBT.read();

        // AUTO MODE
        if (cmd == 'A') {

            autoMode = true;

            Serial.println("AUTO MODE STARTED");
            SerialBT.println("AUTO MODE STARTED");
        }

        // MANUAL MODE
        else if (cmd == 'M') {

            autoMode = false;

            stopMotor();

            Serial.println("MANUAL MODE");
            SerialBT.println("MANUAL MODE");
        }

        // MANUAL CONTROL
        if (!autoMode) {

            switch (cmd) {

                case 'F':
                    forward();
                    delay(50);
                    break;

                case 'B':
                    backward();
                    delay(50);
                    break;

                case 'L':
                    left();
                    delay(50);
                    break;

                case 'R':
                    right();
                    delay(50);
                    break;

                case 'S':
                    stopMotor();
                    delay(50);
                    break;

            }
        }
    }
}

```

```

}

// =====
// AUTONOMOUS MODE
// =====

if (autoMode) {

    int leftDistance = getDistance(TRIG_L, ECHO_L);

    delay(20);

    int rightDistance = getDistance(TRIG_R, ECHO_R);

    delay(20);

    int centerDistance = getDistance(TRIG_C, ECHO_C);

    // CLEAR PATH
    if (centerDistance > 30 &&
        leftDistance > 30 &&
        rightDistance > 30) {

        forward();
    }

    // OBSTACLE DETECTED
    else {

        stopMotor();

        delay(300);

        backward();

        delay(500);

        stopMotor();

        delay(300);

        // CHOOSE BEST SIDE
        if (rightDistance > leftDistance) {

            right();

            delay(700);
        }

        else {

            left();

            delay(700);
        }

        stopMotor();
    }
}

// =====

```

```

// SENSOR DATA EVERY 4 SECONDS
// =====

if (millis() - lastSensorTime >= 4000) {

    lastSensorTime = millis();

    int leftDistance = getDistance(TRIG_L, ECHO_L);

    delay(20);

    int rightDistance = getDistance(TRIG_R, ECHO_R);

    delay(20);

    int centerDistance = getDistance(TRIG_C, ECHO_C);

    int motion = digitalRead(PIR);

    // SCD40
    sendCommand(0xEC05);

    delay(10);

    bool scdStatus = readData(co2, temperature, humidity);

    // =====
    // ALERTS
    // =====

    if (motion == HIGH) {

        digitalWrite(LED, HIGH);

        tone(BUZZER, 1000);

        Serial.println("MOTION DETECTED");
        SerialBT.println("MOTION DETECTED");
    }

    else if (co2 > 1700) {

        digitalWrite(LED, HIGH);

        playTune();

        Serial.println("HIGH CO2 DETECTED");
        SerialBT.println("HIGH CO2 DETECTED");
    }

    else if (centerDistance < 15) {

        digitalWrite(LED, HIGH);

        tone(BUZZER, 2000);

        Serial.println("OBSTACLE DETECTED");
        SerialBT.println("OBSTACLE DETECTED");
    }

    else {

```

```

    digitalWrite(LED, LOW);

    noTone(BUZZER);
}

// =====
// SERIAL OUTPUT
// =====

Serial.println("===== SENSOR DATA =====");

Serial.print("LEFT: ");
Serial.print(leftDistance);
Serial.println(" cm");

Serial.print("CENTER: ");
Serial.print(centerDistance);
Serial.println(" cm");

Serial.print("RIGHT: ");
Serial.print(rightDistance);
Serial.println(" cm");

Serial.print("PIR: ");
Serial.println(motion);

if (scdStatus) {

    Serial.print("CO2: ");
    Serial.print(co2);
    Serial.println(" ppm");

    Serial.print("TEMP: ");
    Serial.print(temperature);
    Serial.println(" C");

    Serial.print("HUMIDITY: ");
    Serial.print(humidity);
    Serial.println(" %");
}

else {

    Serial.println("SCD40 READ FAILED");
    SerialBT.println("SCD40 READ FAILED");
}

// =====
// BLUETOOTH OUTPUT
// =====

SerialBT.println("===== SENSOR DATA =====");

SerialBT.print("LEFT: ");
SerialBT.print(leftDistance);
SerialBT.println(" cm");

SerialBT.print("CENTER: ");
SerialBT.print(centerDistance);
SerialBT.println(" cm");

```

```

SerialBT.print("RIGHT: ");
SerialBT.print(rightDistance);
SerialBT.println(" cm");

SerialBT.print("PIR: ");
SerialBT.println(motion);

if (scdStatus) {

    SerialBT.print("CO2: ");
    SerialBT.print(co2);
    SerialBT.println(" ppm");

    SerialBT.print("TEMP: ");
    SerialBT.print(temperature);
    SerialBT.println(" C");

    SerialBT.print("HUMIDITY: ");
    SerialBT.println(humidity);
}
}
}

```

OBSERVATIONS

Based on testing of the assembled prototype, the following observations were recorded:

- The SCD40 sensor successfully measured CO₂, temperature, and humidity every four seconds over I²C without errors.
- When a person breathed near the sensor in an enclosed test space, CO₂ readings exceeded 1700 ppm, triggering the LED, buzzer, and “HUMAN DETECTED” Bluetooth message.
- Both HC-SR04 sensors correctly measured distances .The front sensor reliably detected obstacles placed within 15 cm.
- The Bluetooth connection was established reliably and movement commands (F,B, L, R, S) were executed with low latency.
- The 2 SLi-Ionbattery pack provided adequate power for continuous operation of all components.

SAMPLE SENSOR READINGS

Table4.2: Representative Sensor Readings During Testing

Test	Condition	CO ₂ (ppm)	Front Dist.(cm)	Temp.(°C)	Status
1	Open room, no human	~480	85	28.3	Nor
2	Human breathing nearby	1700	60	29.1	HUMAN
3	Obstacle placed in front	~490	10	28.5	OBSTACLE
4	Normal operation	~520	45	28.7	Nor

Discussion

The results confirm that the proposed system meets its design objectives.CO₂-based human detection proved effective in enclosed or semi-enclosed test conditions, demon-strating the viability of the SCD40 sensor as a human presence indicator. The dual ultrasonic sensor arrangement provides adequate front and lateral awareness for obstacle detection.

The Bluetooth control interface was reliable over short to medium ranges (tested up to approximately 10 m line-of-sight).The 4-second sensor polling interval represents a trade-off between power consumption, sensor accuracy, and response time that is appropriate for a slow-moving rescue robot.

Minor limitations include the dependence on CO₂ accumulation in enclosed spaces, which may reduce effectiveness in open-air outdoor scenarios .Future integration of additional sensors (PIR, thermal camera) and autonomous navigation algorithms would further improve the system.

Chapter5

Conclusion and Future Scope

Conclusion

This project successfully designed, implemented, and tested an IoT-based autonomous disaster rescue robot capable of detecting human presence and avoiding obstacles in hazardous environments. The system integrates an SCD40CO₂ sensor and dual HC-SR04 ultrasonic sensors with an ESP32 WROOM microcontroller, mounted on a four-wheel smart chassis driven by an L298 motor driver.

The key conclusions of this work are:

- CO₂-based human detection using the SCD40 sensor is effective in enclosed or semi-enclosed environments, successfully identifying human presence when CO₂ concentration exceeds 1700 ppm.
- Dual HC-SR04 ultrasonic sensors provide reliable front and lateral obstacle detection, correctly triggering alerts when obstacles are within 15 cm.
- The ESP32 Bluetooth module enables low-latency remote control and real-time sensor data transmission, providing continuous situational awareness to rescue operators.
- The complete system was implemented using affordable, commercially available components, demonstrating the viability of a low cost rescue robot platform.
- The prototype operated reliably during testing, confirming that the proposed approach meets its design objectives.

Overall, this project establishes a practical and affordable foundation for IoT-enabled rescue robotics, with demonstrated capability for human detection, obstacle avoidance ,and wireless operator communication.

Future Scope

Although the proposed system demonstrates promising outcomes, there remains significant scope for future improvements and extensions. Some possible directions for future work include:

Additional Sensors: Integration of a thermal camera to complement CO₂-based detection and improve accuracy across open and enclosed environments.

Wi-Fi/Cloud Integration: Replacing or supplementing Bluetooth with Wi-Fi or LoRa for longer-range communication and cloud-based data logging and monitoring.

Camera Integration: Adding a live video feed (e.g., OV2640 camera module) for visual situational awareness.

Extended Battery Life: Optimization of power management and use of higher capacity battery packs for prolonged field operation.

Robust Chassis Design: Upgrading to a tracked or more ruggedized chassis suitable for uneven terrain typical of disaster sites.

Mobile Application: Development of a dedicated Android/iOS application for intuitive remote control, sensor data visualization, and alert management.

These directions provide clear pathways to evolve this prototype into a more capable and deployable rescue robotic system suitable for real-world disaster response operations.

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